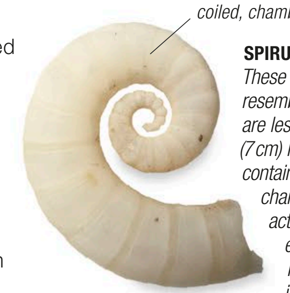
**Class** Cephalopoda

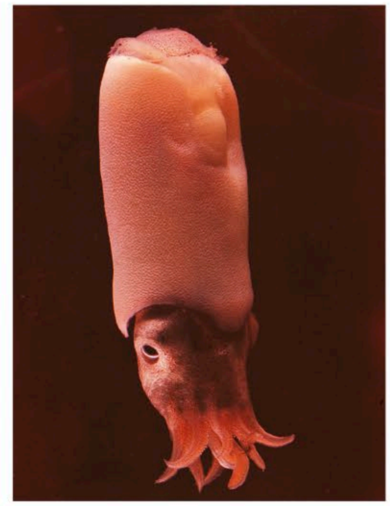
# Cephalopods

**Occurrence** About 750 spp. worldwide; marine, in open water and on or near seabed

Cephalopods include the largest, fastest, and most intelligent of all invertebrates. Unlike other mollusks, they have arms and tentacles (modified arms), and they can move with speed by a form of jet propulsion, which involves squirting water out of the mantle cavity. The arms and tentacles are used to catch prey. In many species, other modified arms are used by males to transfer spermatophores (sperm packets) to the female. Cephalopods have a rasping mouthpart (radula), a birdlike beak, a large brain, and well-developed eyes. Many early cephalopods had coiled, external shells, but among the existing species, only the nautiluses show this characteristic. In all other cephalopods, such as cuttlefishes, squids, and octopuses, the shell is either internal or absent entirely. Cephalopods vary greatly in size: from tiny *Idiosepius* species, that are just ⅜–¾ in (1–2 cm) long, to giant and colossal squids. Many cephalopods breed only once and then die.



*coiled, chambered shell*

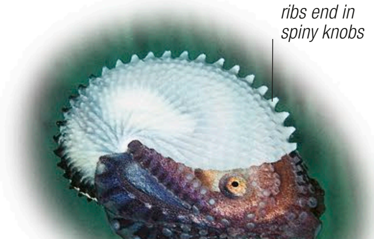


**SPIRULAS**
*These small creatures resembling cuttlefishes are less than 2¾ in (7 cm) long. Their bodies contain a coiled and chambered shell that acts as a float, enabling them to hang head down in the water (see right), at depths of up to 6,600 ft (2,000 m). Using their small arms, they catch planktonic animals that drift within reach. When they die, their shells can float to the surface, and are often carried onto the shore. Shells of the common spirula,* Spirula spirula, *shown, are found on tropical beaches.*

**NAUTILUSES**
*Nautiluses are the only surviving cephalopods with a true external shell, which may be up to 8 in (20 cm) across. It contains a series of gas-filled chambers that act as a buoyancy device, and a large final chamber that houses the animal itself. The pearly nautilus,* Nautilus pompilius *(above), lives in the Indo-Pacific.*

**GIANT SQUIDS**
*Members of the genus* Architeuthis, *distributed worldwide, giant squids can be over 58 ft (17⅗ m) long. No giant squids has ever been captured alive but they have been photographed. Their bodies are found on beaches, and their undigested beaks in the guts of sperm whales.*

*ribs end in spiny knobs*



**PAPER NAUTILUS**
*A species in the genus* Argonauta, *the pantropical paper nautilus (shown above) is really a type of octopus. It is 8 in (20 cm) long. The breeding female has a superficial, nautilus-like, thin shell—a case secreted by 2 of the arms—in which she broods the fertilized eggs.*

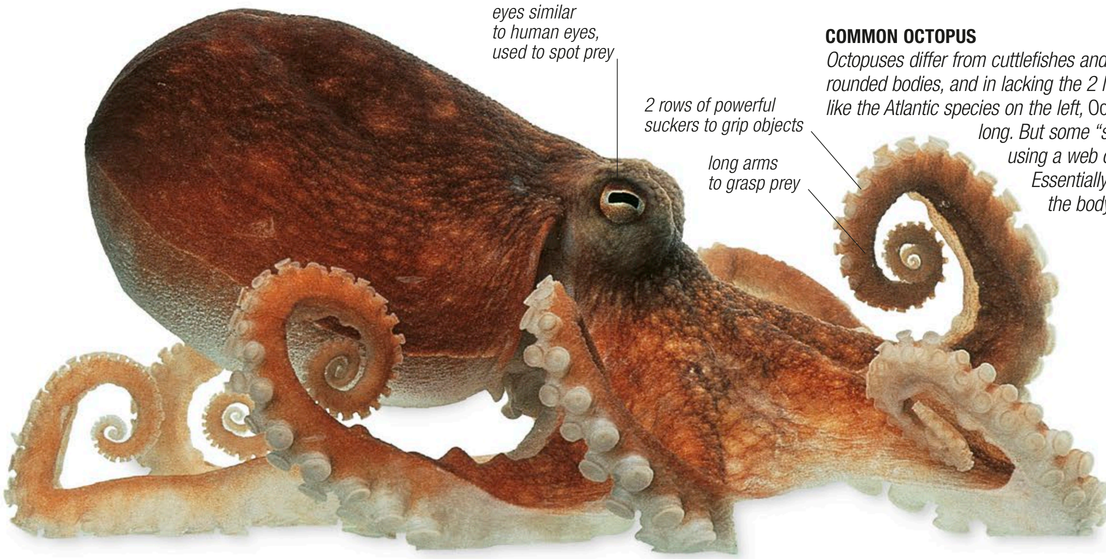
**PACIFIC SQUIDS**
*Squids are torpedo-shaped hunters, with internal shells that are reduced to a small, horny "pen." Some hunt alone, but many medium-sized species—such as the Pacific* Doryteuthis opalescens, *above, up to 8 in (20 cm) long—feed in large schools. All swim by flapping their fins, or by jet propulsion. Squids have 2 long, prey-catching tentacles that can be retracted, but not completely as in cuttlefishes.*

*undulating fins along sides*

**CUTTLEFISHES**
*Cuttlefishes have a flat, internal shell—the familiar "cuttlebone" that is often washed up on the shore. Their bodies are short and broad, with 8 short arms and 2 much longer tentacles that can shoot out to catch prey. Cuttlefishes are known for their ability to change color. The species shown here,* Sepia officinalis, *is common around the eastern North Atlantic and grows up to 12 in (30 cm) long.*

**HELICOCRANCHIA SP.**
*This small, semitransparent piglet squid is found in deep water between 328 and 656 ft (100–200 m) in many of the world's oceans. It has large light-producing organs below each of its eyes, but little is known about its behavior.*

**BLUE-RINGED OCTOPUS**
*The blue-ringed octopus,* Hapalochlaena lunulata, *inhabits coral reefs in the Pacific and Indian oceans. It is small—4 in (10 cm) long—and beautiful, but has a venomous bite that can kill humans in 15 minutes.*



*eyes similar to human eyes, used to spot prey*

*2 rows of powerful suckers to grip objects*

*long arms to grasp prey*

**COMMON OCTOPUS**
*Octopuses differ from cuttlefishes and squids in having more rounded bodies, and in lacking the 2 long, prey-capturing tentacles, like the Atlantic species on the left,* Octopus vulgaris, *3¼ ft (1 m) long. But some "sail" bottom currents using a web of skin between the arms. Essentially they have no shell and the body is incredibly flexible.*

**WUNDERPUS PHOTOGENICUS**
*This octopus of Indo-Pacific shallow waters was described in 2006. It has small, stalked eyes and a distinctive white-spotted pattern that becomes more pronounced when the animal is disturbed. It preys on fishes and invertebrates at twilight.*